Exercise 1:

a) The forcast origin is $t = 2007$, we have:

$$\hat{Y}_t(1) = E[Y_{t+1} \mid Y_t, Y_{t-1}, Y_{t-2}]$$

$$= 1,1\, Y_t - 0,5\, Y_{t-1} + 5$$

$$= 1,1(10) - 0,5(11) + 5$$

$$\boxed{\hat{Y}_t(1) = \hat{Y}_{2008} = \$10,5 \text{ million}}$$

$$\hat{Y}_t(2) = \hat{Y}_{2009} = 1,1\, \hat{Y}_t(1) - 0,5\, Y_t + 5$$

$$= 1,1(10,5) - 0,5(10) + 5$$

$$\boxed{\hat{Y}_t(2) = \hat{Y}_{2009} = \$11,55 \text{ million}}$$

b) We have: $\psi_0 = 1$ and $\psi_1 = \phi_1 \psi_0$ which yields

$$\boxed{\psi_1 = 1,1}$$

c) A 95% prediction limit for 2008 assuming $e_t \sim \mathcal{N}(0, \sigma_e^2)$ is:

$$10,5 \pm 1,96\sqrt{2} = \boxed{10,5 \pm 2,7}$$

d)

$$\hat{y}_{t+1}(1) = \hat{y}_{2009} = \hat{y}_t(2) + \varphi_1(y_{t+1} - \hat{y}_t(1))$$

$$= 11,55 + 1,1(12 - 10,5)$$

$$\hat{y}_{t+1}(1) = \hat{y}_{2009} = \$\,13,2 \text{ million}$$